



FIRST SEMESTER 2026-2027

Course Handout (Part II)

August 01, 2026

In addition to part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course:

Course Number & Title : **CS/SS G 527 Cloud Computing**
Instructor In-Charge : **HARI BABU K**
Course Website : **Nalanda**

1. Scope and Objective of the Course

The primary objective of the course is to introduce the student to cloud computing from architectural and design perspectives. As such, the emphasis of the course would be on the underlying infrastructure and architecture of clouds, techniques for enabling services and the quality of such services, as well as issues in designing clouds. Select research issues in performance, security, and management would also be addressed. Programming on the cloud would be encouraged but not taught in class. Students are expected to learn and understand tools and techniques for using, designing, and implementing clouds and services via assignments and a term project.

2. Text Book

T1. None

3. Reference Books

- R1. Kai Hwang , Jack Dongarra , Geoffrey C. Fox Distributed and Cloud Computing: From Parallel Processing to the Internet of Things. Morgan Kauffman 2011
- R2. Jim Smith, Ravi Nair. Virtual Machines: Versatile Platforms for Systems and Processes. Morgan Kaufmann. 2005
- R3. [Fox, Armando, et al. "Above the clouds: A berkeley view of cloud computing." Dept. Electrical Eng. and Comput. Sciences, University of California, Berkeley, Rep. UCB/EECS 28.13 \(2009\): 2009.](#)
- R4. Thomas, Erl, Mahmood Zaigham, and Puttini Ricardo. "Cloud Computing Concepts, Technology & Architecture." (2013).
- R5. Alex Xu, System Design Interview - An insider's guide, Second Edition, 2020.
- R6. [Grattafiori, Aaron. "Understanding and hardening linux containers." Whitepaper, NCC Group \(2016\).](#)
- R7. [Barroso, Luiz André, and Urs Hölzle. "The datacenter as a computer: An introduction to the design of warehouse- scale machines." Synthesis lectures on computer architecture 4.1 \(2009\): 1-108.](#)
- R8. Kleppmann, Martin. Designing data-intensive applications: The big ideas behind reliable, scalable, and maintainable systems. " O'Reilly Media, Inc.", 2017.

AR: Additional readings to be provided



4. Course Plan:

a. Modules

Module	Theme	Learning Objectives
I	Cloud Computing – Introduction	• Learn how computing evolved over decades • Learn cloud computing definition and its defining characteristics. • Learn the service delivery models, deployment models • Learn economic aspects of Cloud Computing • Economic models and SLAs
II	Cloud Infrastructure	• Learn the hardware building blocks of a cloud datacenter (compute, storage and network) • Learn energy and power efficiency of cloud datacenter • Learn workloads and software infrastructure in a cloud datacenter
III	Virtualization	• Learn about process VMs and System VMs • Learn the approaches to virtualize (classical, software-assisted, para-virtualization, hardware-assisted) • Learn how to virtualize CPU, Memory, and I/O in all three approaches using case studies (VMWare, Xen, Intel VT-x) • Learn about Linux Containers • Learn about Namespaces, Control Groups, Capabilities • Explore LXC and Docker
IV	Cloud Storage	• Learn about storage models, file systems and databases • Learn cloud storage concepts (consistency, replication, consensus) • Learn about large scale distributed file systems (GFS, HDFS, Ceph FS) • Learn about cloud databases (Dynamo DB) and explore HBase, MongoDB, Cassandra. • Learn about cloud object storage (Amazon S3) and explore OpenStack Swift, Ceph.
V	Programming Models	• Learn the concepts of distributed programming for the Cloud • Learn batch processing programming model (Hadoop MapReduce) • Learn programming model for iterating and interactive applications (Spark) • Learn programming model for stream processing (Storm)



VI	Cloud Management, Performance and Security Issues	• Learn about SLAs, SLOs, availability, scalability, and elasticity • Learn the principles and techniques in scaling services and building them • Learn about software defined networking and storage in Cloud Data centers (Openstack) • To understand performance issues and techniques to enable performance of a cloud at different levels of abstraction • To understand security issues specific to cloud computing and solutions to address them.
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b. Lecture Schedule:

Lectures	Module	Topic	Reference
1-2	I	Evolution of Computing Systems – Mainframes to PCs to Networked Systems to Clouds. Introduction to Cloud Computing – NIST definition, enablers and economies of scale.	R3: Sec 1 to 7
3	I	Introduction to Cloud Computing – product vs process, service orientation of a process, computing as a utility.	Class Notes + R1: Sec 1.4.1
4	I	Introduction to Cloud Computing – Service delivery models, cloud deployment models with examples.	R4: Ch 14
5	VI	Service Level Agreements (SLAs), Service Level Objectives (SLOs); Availability: high availability models, measurement;	R4: Ch 16 R1: 2.3.2-3
6-7	VI	Scalability: Principles and techniques; Scaling services: Web services; Micro services; Elasticity;	R5: Ch1 + AR
8	V	Programming models for the cloud	AR
9	V	Programming on Clusters: Map-reduce architecture;	AR
10	V	In-memory Distributed Programming: Spark; Stream processing on cluster: Storm.	AR
11	V	Server-less computing paradigm;	AR
12	III	Virtualization: Introduction, Different Levels of Abstraction, and Mechanisms for Virtualization. Process vs. System VMs, HLL VMs.	R2: Ch1
13-14	III	Emulation – Interpretation; Binary translation;	R2: Section 2.1-3,2.5
15	III	System VMs: ISA Virtualizability and Resource Virtualization – CPU	R2: Section 8.1-2
16	III	System VMs: Virtualization of Memory (VMWare) and I/O	R2: Section 8.3-4
17	III	Para virtualization: Xen; Hardware-assisted virtualization: Intel VT	R2: Section 8.7



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18	III	Containers: Namespaces, Control Groups, Capabilities;	R6: Ch 1-5
19-20	II	Hardware building blocks of a cloud data center: compute, storage, network; challenges in storage and networking;	R7: Ch1 & Ch3
20	II, VI	Workloads and software infrastructure in a cloud datacenter	R7: Ch2
21-22	II, VI	Network virtualization: Datacenter network topologies; Routing solutions; SDN;	R1: Sec 3.4 R2: Sec 10.2 AR
23-24	II, VI	Storage virtualization: NAS and SAN, Virtual Storage; SDN	AR
25	VI	Resource management: Live migration	AR
26	VI	Resource management: Monitoring	AR
27-28	VI	Resource management: Global resource management (Borg)	AR
29-30	VI	Resource management: Local resource management (VMWare)	AR
31	IV	Distributed storage: file systems, databases, challenges in scaling, challenges in distributed storage;	R8: Ch 8
32-33	IV	Distributed storage: replication vs partitioning, leaderless replication; Dynamo; S3;	R8: Ch 5 + AR
34	IV	Data partitioning: approaches, balancing partitions, request routing;	R8: Ch6
35-36	IV	Data replication: consistency models, consensus through 2PC, 3PC, Paxos, active vs passive replication;	R8: Ch 9 + AR
37	IV	Consensus in the cloud: ZAB, raft protocols; Zookeeper	AR
38-39	IV	Large scale distributed file systems: GFS, Ceph FS	AR
40	IV	Cloud databases: Cap theorem, PACELC theorem; NoSQL Databases;	AR
41-42	VI	Security issues specific to cloud computing and solutions to address them.	AR

5. Practice Sessions

Following practice sessions are planned during the semester.

- Practice Session1: Load Balancing and Auto Scaling on AWS
- Practice Session2: Infrastructure as Code (IaC)
- Practice Session3: AWS Elastic Mapreduce (EMR) - Hadoop, Spark
- Practice Session4: Docker Containers
- Practice Session5: Microservices



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- Practice Session6: DevOps and CI/CD
- Practice Session7: Virtual Machine Management
- Practice Session8: Apache ZooKeeper
- Practice Session9: Kubernetes
- Practice Session 10: Kubernetes

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date, Time & Venue	Remarks
Mid-Semester Test	90 Mins	25%	As per Timetable	Partial Open book
Take Home Exercises based on labs		10%	TBA	Take Home
Three quizzes based on labs		10%	TBA	Best two will be considered. Closed book.
Assignment		15%	TBA	Programming, Take Home
Comprehensive Examination	3 Hours	40%	As per Timetable	Partial Open book

7. Notices:

All notices shall be displayed only on course webpage.

8. Malpractices:

While coding assignments/lab exercises you are not allowed to share source code but discussions are allowed with others. Any copying detected among groups/individuals will be reported to appropriate authority.

9. Make-up Policy:

No makeup will be given for Labs and Assignment components. For tests, however, make-up will be granted strictly on prior permission and on justifiable grounds only.

10. Chamber Consultation Hours:

Thu 4-5PM in 6121-M

Instructor-in-charge CS/SS G527



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